BayWa r.e.

munich climate week

BAYWA R.E. CHALLENGE · MUNICH CLIMATE WEEK 2026

Estimating lost energy after a wind park outage

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1 SEPTEMBER 2026, 00:00

Wind park T was offline for 72 hours. How much energy did it lose?

SITUATION & TASK

1–3 September: wind park T goes silent for 72 hours.

STILL AVAILABLE

Wind measurements from seven reference turbines in two nearby parks, H and O.

MISSING

The wind speed at turbines T1, T2 and T3: 432 ten-minute values each.

OUR TASK: A TOOL THAT

Detects the outage → Estimates the wind → Calculates kWh lost → Stays explainable

ACTION · THE IDEA

Nearby wind parks feel the same weather.

1. The same weather systems pass over parks H, O and T, so their wind rises and falls together.
2. We measure how closely each reference turbine follows each T turbine: the correlation, from 0 (unrelated) to 1 (identical).
3. The model combines all seven reference turbines, each with its own weight. Combined, they follow T better than any single one: 0.85 instead of 0.80 for T2.

REFERENCE TURBINES · CORRELATION

	T1	T2	T3
O1	0.73	0.80	0.77
O2	0.71	0.80	0.77
H5	0.69	0.77	0.74
H3	0.68	0.77	0.74
H1	0.70	0.72	0.70
H4	0.67	0.74	0.73
H2	0.67	0.73	0.72

Park O is the closest match for all three turbines.

ACTION · THE METHOD

From reference wind to lost kWh

Outage detection

Longest block where T1–T3 are all missing: 01.09 00:00 – 03.09 23:50. Gaps are never read as zero wind.

Calibration

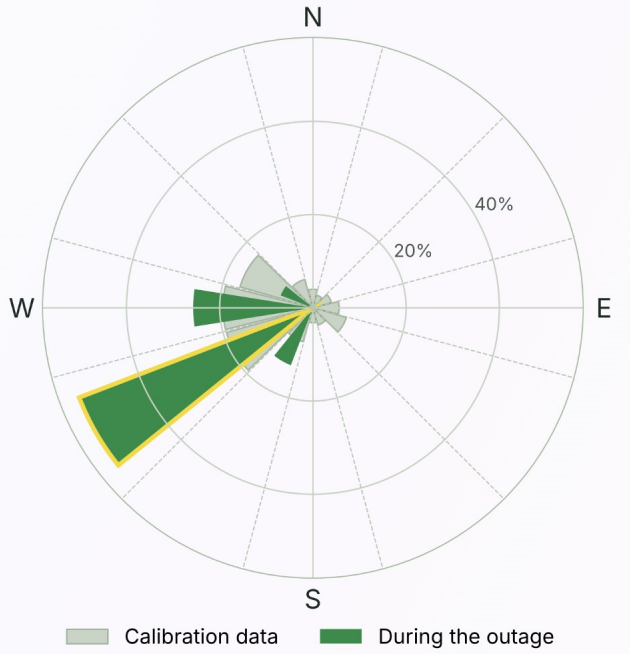
The wind direction at park H sorts every 10 minutes into one of 12 sectors of 30°. Per sector and turbine: a linear regression on all seven reference turbines.

Reconstruction

During the outage, each 10-minute value uses the equation of the sector the wind came from.

Energy calculation

Power from the manufacturer's power curve, × 10/60 h per interval, summed over 432 intervals.



Example · sector 225–255°, 54% of the outage

T1 = 0.10 + 0.35·01 + 0.28·H2 + 0.14·H5 + 0.12·H4 − 0.05·H1 + …

Steps as in the tool's Methodology tab · tested against an alternative model with equal accuracy

RESULT · 1–3 SEPTEMBER

ENERGY LOSS

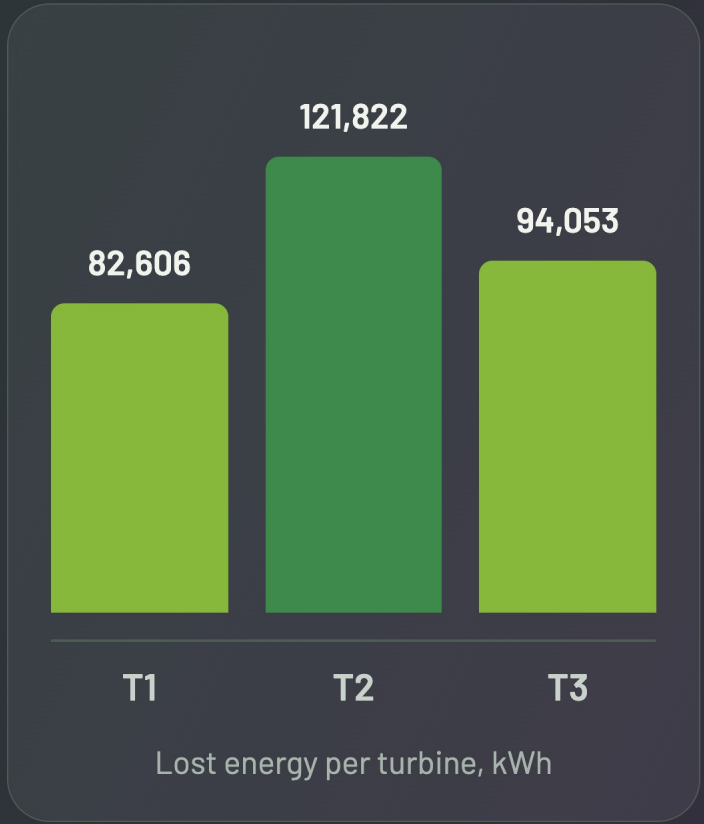
298,481 kWh

Expected range 268,000 – 329,000 kWh

REVENUE LOSS

≈ €32,700

At 10.94 ct/kWh, before policy terms



Price: EEG market value for onshore wind, September 2026 (netztransparenz.de), confirmed as typical by BayWa mentors

RESULT · ACCURACY

Checked against measured values

Wind park T	Training + validation · Jul–Aug	Blind test	Test
Reference parks H, O	Training + validation · Jul–Aug	1–3 Sep	4–9 Sep
		↑ energy loss	↑ RMSE, MAE

Test days 4–9 September	T1	T2	T3
Wind speed MAE (m/s)	1.14	0.96	1.03
Wind speed RMSE (m/s)	1.47	1.27	1.35
Power MAE (kW)	618	532	618
Power RMSE (kW)	880	790	867
Energy deviation	+10.6%	−0.4%	−4.3%

28 SIMULATED OUTAGES

±10%

Typical deviation of the energy total per three-day outage, slightly on the low side.

Data split as in the challenge briefing. MAE and RMSE per 10-minute value.

RESULT · LIMITATIONS

What the model does not know

Distance and terrain

Not provided. The relationship between the parks is learned from the data alone.

Season and year

Only three summer months. Winter storms, icing and other years may behave differently.

Normal operation

Assumed during the outage: no maintenance, faults or curtailment by the grid operator.

Power curve as specified

No correction for air density, ageing or icing of the turbines.

Monthly price

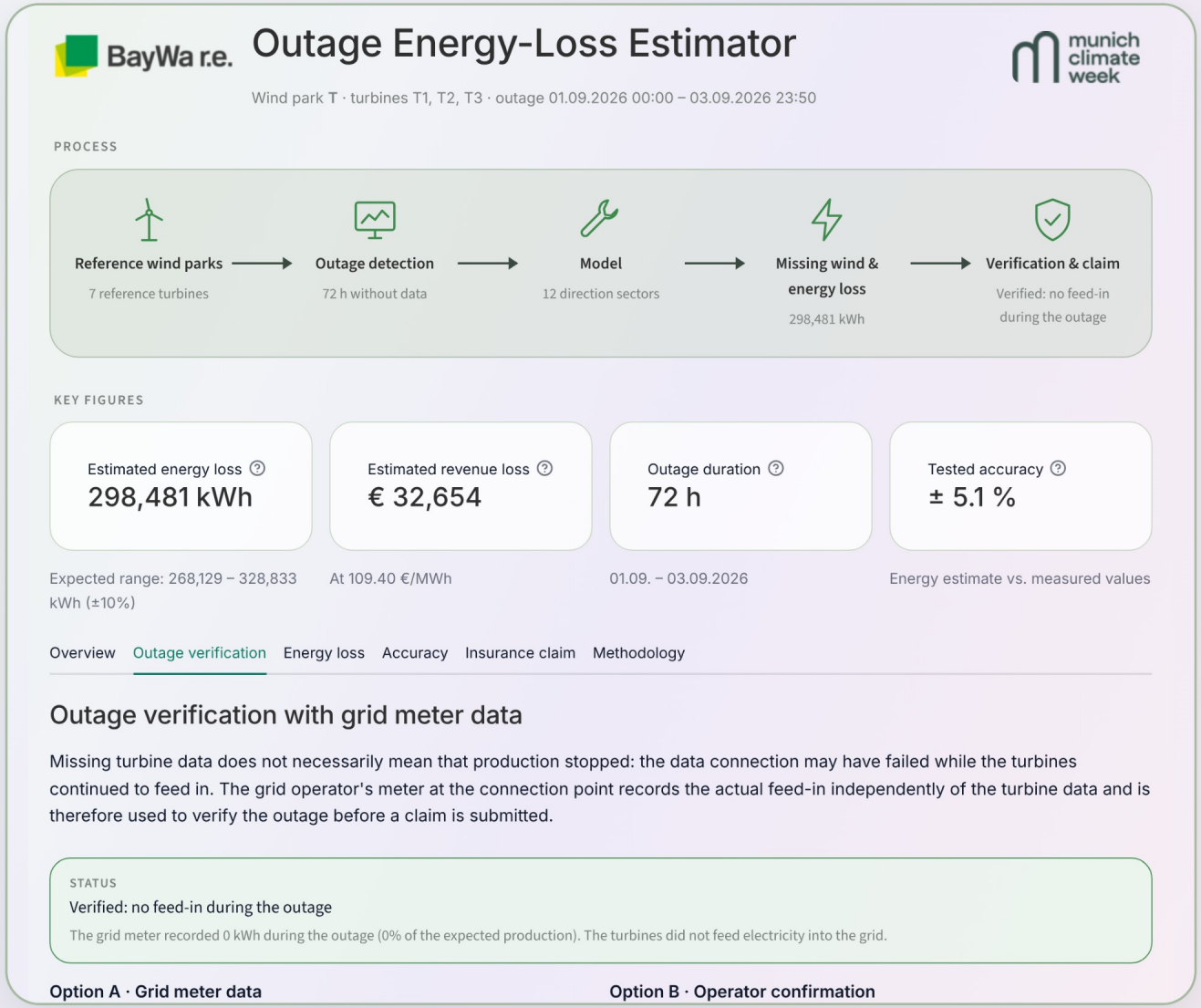
Revenue uses the monthly market value, not hourly prices or the farm's own contract.

RESULT · THE TOOL

Built for the person who files the claim.

- Detects the outage automatically
- Verifies it against grid meter data
- Shows loss, accuracy and method
- Drafts the insurance notification

Runs locally in the browser. CSV and Excel upload, German formats included.



THE ANSWER

How much energy did it lose? 298,481 kWh.

CERTAINTY

±10%

Tested, slightly conservative

REVENUE

≈ €32,700

Before policy terms

PROOF

Verified

Against grid meter data

Next: an automatic reliability check per claim, more seasons of data, hourly prices and curtailment periods.

Thank you.

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